

Yucel Yavuz

Assoc. Director, AI & Data Science | GenAI | ML | Product Strategy

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SUMMARY

AI & Data Science leader with 7+ years of hands-on experience in machine learning, Generative AI, and AI product development, backed by 10+ years of cross-functional leadership. Experienced in architecting and deploying end-to-end AI solutions using Databricks, Snowflake, Python, and modern LLM frameworks, including predictive modeling, explainable AI, and GenAI-powered automation. Strong track record of translating complex business opportunities into scalable AI products and measurable business impact through effective collaboration with technical and non-technical stakeholders.

TECHNICAL & SOFT SKILLS

AI, Machine Learning & Analytics: Generative AI • Agentic AI • LangChain • LangGraph • RAG • LLM Applications • Explainable AI (SHAP) • Predictive Modeling • Customer Segmentation • Causal Inference • A/B Testing • Time Series Forecasting • NLP

Platforms/Engineering: Databricks • Snowflake • Python • PySpark • SQL • Streamlit • MLflow • Docker • Git/GitHub • CI/CD • Databricks AI • Snowflake Cortex

AI Product & Architecture: AI Product Development • End-to-End AI Deployment • Production ML Pipelines • AI Solution Architecture • Workflow Automation • Decision Intelligence

Visualization & BI: Plotly • Tableau • Power BI • Automated Dashboarding

Leadership: Cross-Functional Leadership • AI Strategy • Stakeholder Management • Team Mentorship • Business Translation • Executive Communication

PROFESSIONAL EXPERIENCE

Horizon Media – New York, NY

Assoc. Director, AI & Data Science | Generative AI | Apr 2023 – Present

- **Architected** LLM-driven analytics solutions (LangChain, Streamlit, Snowflake Cortex) that transformed demographic and behavioral data into actionable decision intelligence, accelerating stakeholder decision-making by 35% and reducing manual reporting by 40%.
- **Led development of** explainable AI frameworks on Databricks using SHAP, increasing model transparency and stakeholder adoption by 20% across analytics workflows.
- **Designed and operationalized** Databricks–Snowflake ML pipelines for LTV and look-alike modeling using XGBoost and ensemble methods, achieving 92% model accuracy and improving campaign efficiency by 15%.
- **Built** scalable audience intelligence and growth frameworks integrating behavioral, transactional, and temporal data to uncover key conversion drivers, improving customer lifecycle performance by ~10% through personalized targeting strategies.
- **Applied** causal inference methodologies (A/B Testing, Propensity Score Matching) to quantify campaign lift and ROI, improving targeting performance by ~12% and strengthening media investment decisions.
- **Delivered** ML- and GenAI-driven audience profiling frameworks that refined audience strategies, strengthened new business pitches, and contributed to two major client wins.
- **Directed** audience segmentation and behavioral analytics initiatives in Databricks, identifying high-value customer segments that increased retention by 12%.
- **Designed** interactive Tableau and Power BI dashboards visualizing 1P metrics, cross-sell trends, and behavioral clusters, reducing insight turnaround by 30% and enabling real-time strategic planning.

Bowery Farming – Kearny, NJ

Data Scientist | Operations Manager | Sep 2018 – Apr 2023

- Developed predictive yield models comparing **7 ML and 1 DL algorithm**, increasing yield accuracy **by 10%** and improving production planning.
- Built automated **production dashboards**, enhancing productivity **by 25%** and warehouse efficiency **by 20%**.
- Introduced machine learning-driven process optimization that **reduced packaging losses by 20%**.
- Created a Team Lead Development Tool and training program that **promoted 6 employees, reducing turnover to 2%**.
- Created **a great team culture where employees worked efficiently** while ensuring production compliance with company quality, sanitation, and food safety standards [2019 and 2020 SQF (Safe Quality Food) results were 98% and 2021 SQF result was 99% - Excellent].

Teleplan, Mechanicsburg, PA

Sr. Analyst | Supply Chain | Jan 2017 – Sep 2018

- Developed a **supply chain analytics dashboard** improving efficiency by **\$150K** in 8 months and supplier performance to **92.6%**.
- Streamlined forecasting, procurement, and vendor workflows, achieving **99.8%** cycle count accuracy.
- Introduced analytical process improvements that optimized material reuse and cost savings initiatives.

Turkish Army, Ankara/Turkey

Operations and Analytics Manager | Aug 2001-April 2016

- Spearheaded **10+ strategic analytics projects**, saving **\$300K** in resource allocation.
- Designed regression-based forecasting models for personnel and logistics planning.
- Delivered **data-driven reports and dashboards** for executive decision-making across multiple departments.

RECENT PROJECTS

- **AI Media Planning Agent – Multi-Agent LLM System for Marketing Strategy**

Python, LangGraph, LangChain, OpenAI, Streamlit

Architected a multi-agent AI platform that automates end-to-end marketing workflows, transforming natural language prompts into business insights, audience strategy, media plans, creative recommendations, and personalized communications.

- **Business Intelligence Chatbot – Text-to-SQL AI Platform (Databricks)**

Python, Streamlit, Databricks AI (ChatDatabricks), LangChain, LangGraph, Plotly

Designed and deployed a conversational BI platform on Databricks that translates natural language into SQL queries, executes analytics workflows, and delivers interactive visual insights through a self-service Streamlit application.

- **SHAP Analysis Workflow – End-to-End Explainable AI Product on Databricks**

Python, XGBoost, SHAP, Streamlit, Databricks AI, Databricks Feature Store, Snowflake

Architected an end-to-end explainable AI product combining XGBoost modeling, SHAP driver analysis, and GenAI-powered executive reporting to deliver scalable stakeholder decision intelligence.

- **Affinity Engine – AI-Powered Product Affinity Platform**

Python, Streamlit, Databricks AI, Snowflake, Plotly

Architected an AI product that analyzes large-scale customer transaction and audience data to identify product affinity, cross-sell opportunities, and audience expansion strategies for media and growth planning.

- **Cluster Analysis Workflow – Audience Segmentation (GenAI) Powered on Databricks**

Python, PySpark MLlib, Databricks AI, Snowflake

Designed an automated audience segmentation pipeline leveraging PCA, clustering, and GenAI-powered executive reporting to generate customer personas, behavioral insights, and activation strategies.

- **Marketing Assistant CoPilot – RAG-Based Knowledge Assistant**

Python, Streamlit, LangChain, OpenAI, FAISS

Built a RAG-powered knowledge assistant enabling users to summarize and query enterprise documents via natural language, delivering contextual insights and executive-ready summaries.

EDUCATION

- **U.S. Army War College – Carlisle, PA**

International Fellows Academic Prep Course

- **Turkish National Defense University – Ankara, Turkey**

Ph.D. (Pending), Technology Management

M.S., Technology Management

B.S., Systems Engineering

- **Turkish Army War College – Istanbul, Turkey**

M.A., Organizational Management & Leadership

CERTIFICATIONS

- Generative AI for Data Scientists – *IBM*
- Generative AI Bootcamp – *Business Science*
- Master Machine Learning for Data Science & AI - *CodeBasics*
- Databricks Mosaic AI – *Coursera*
- Certified Management Executive (CME) – *Management & Strategy Institute*